

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SOLUTIONS

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Aim:

To analyze ARP Request and ARP Reply packets and understand how IP addresses are resolved into MAC addresses.

Software Used:

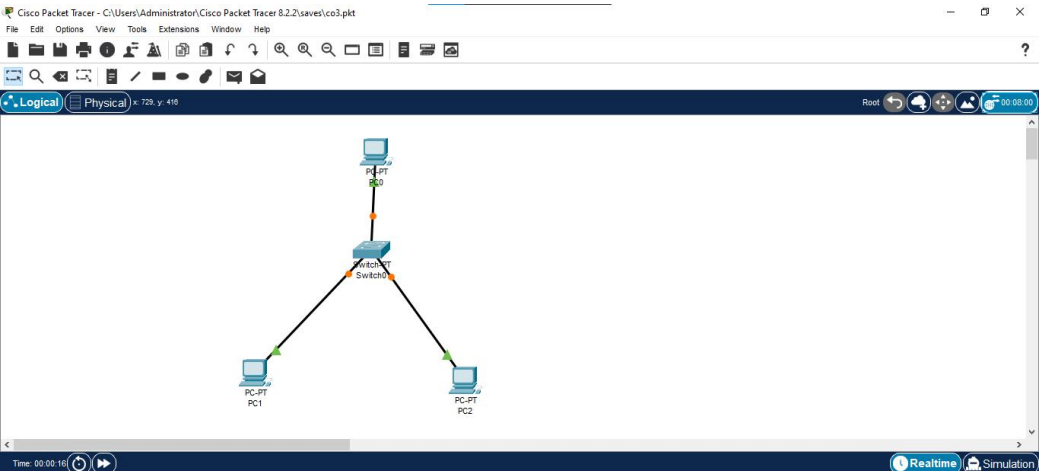
Cisco Packet Tracer and Wireshark

Procedure:

- 1. Open Cisco Packet Tracer and create a new workspace.
- 2. Add three PCs and one switch and connect them using Ethernet cables.
- 3. Configure suitable IPv4 addresses for all three PCs in the same network.
- 4. Check and clear the ARP cache of PC0.
- 5. Open Wireshark and apply the ARP filter.
- 6. From PC0, ping another PC using its IP address.
- 7. Capture and identify the ARP Request and ARP Reply packets.
- 8. Analyse the sender and target IP/MAC addresses and verify the ARP resolution.

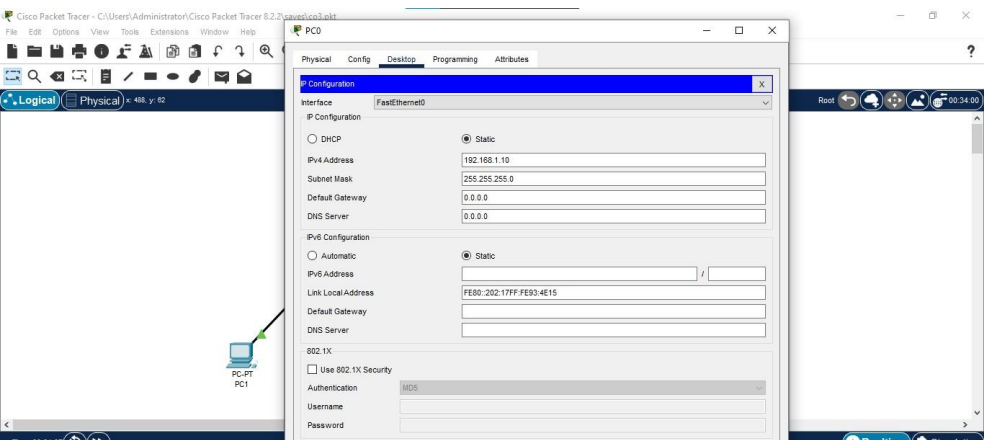
Create the network:

Connect the 3 PCs with the switch with the copper straight cable and create the network of this structure.



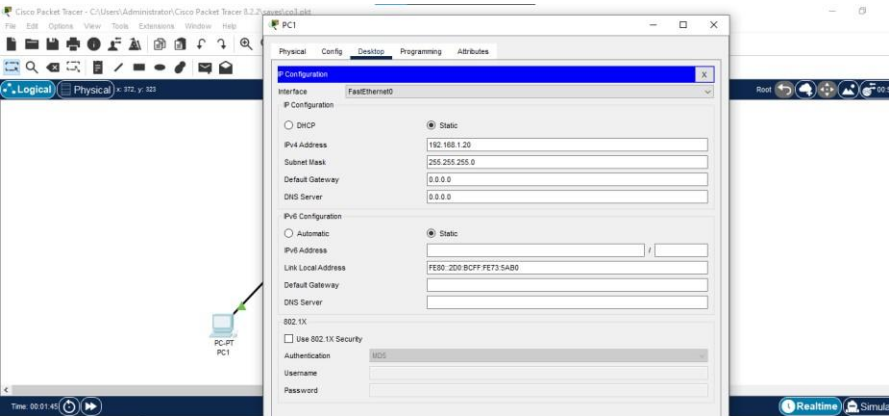
Configure PC0:

For assuming the IP addresses click the PC0 and go to desktop and click the IP configuration then set the address.



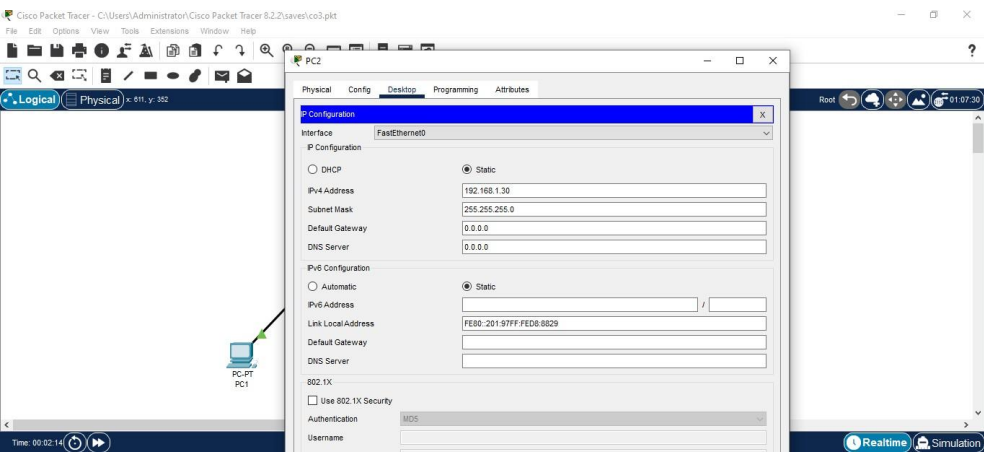
Configure PC1:

For assuming the IP addresses click the PC1 and go to desktop and click the IP configuration then set the address.



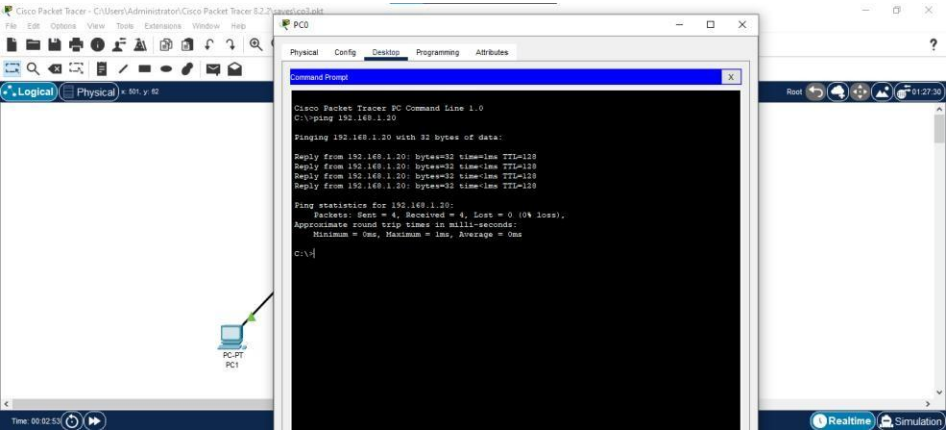
Configure PC2:

For assuming the IP addresses click the PC2 and go to desktop and click the IP configuration then set the address.



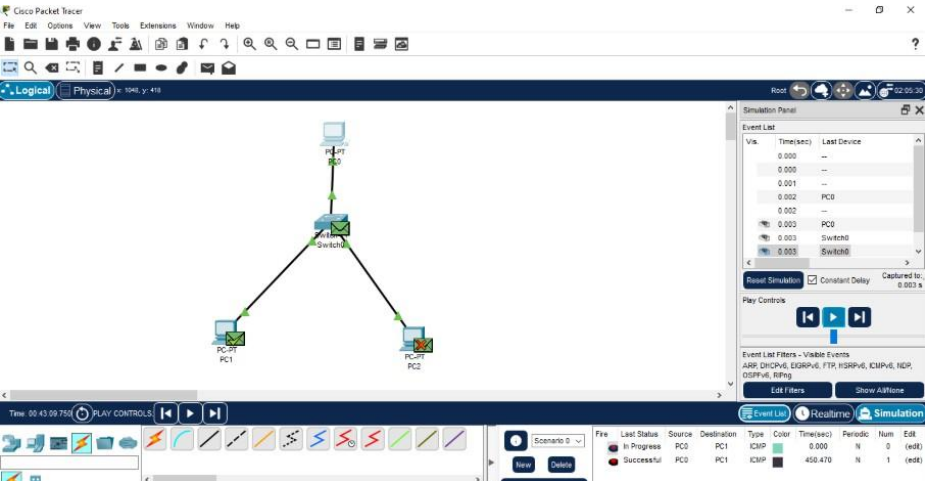
Pinging the IP addresses:

To check whether the IP of one device is replying on another device we have to click the PC0 and go to command prompt then give the command ping by continuing the IP of PC1.



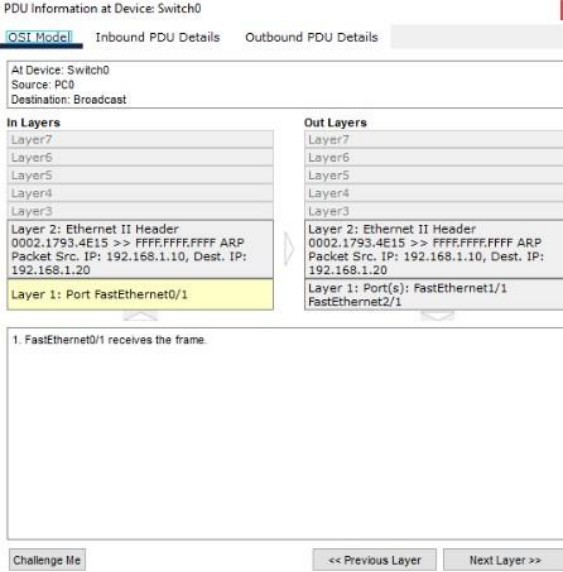
Packet transfer:

By giving the message packets passing from PC0 to PC1 check whether the data transfer is successful or not.



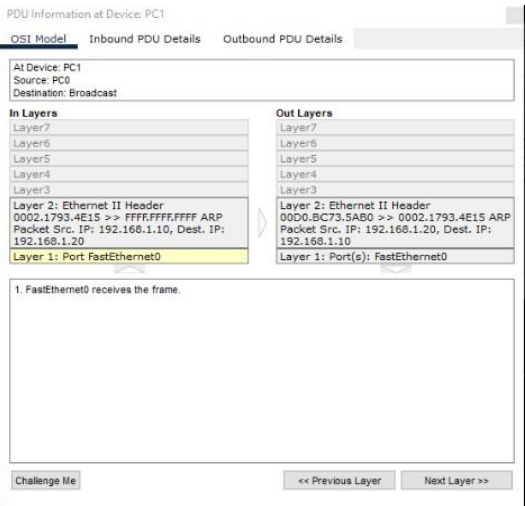
ARP request:

As we have given to find the ARP request to send and receive the reply so we will change the form to ARP and first give request from PC0 to PC1.



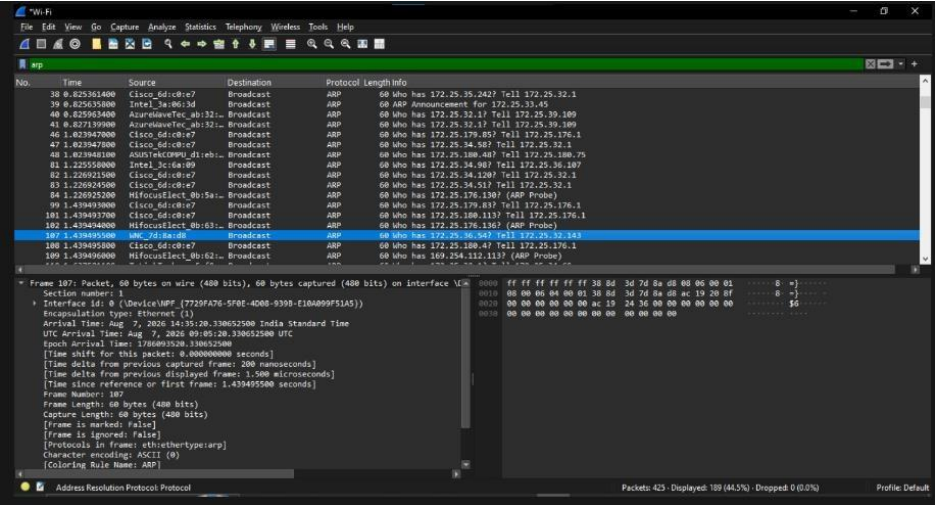
ARP reply:

After giving the request, we got the reply from PC0 to PC1 that source to destination.



Wireshark:

This will be the comparison of ARP values in wireshark by conquering it with the cisco packets.



Conclusion:

Thus, the ARP Request and ARP Reply were successfully analyzed using Cisco Packet Tracer Simulation Mode, while Wireshark was used to observe ARP traffic on the active network interface. ARP successfully resolves a logical IPv4 address into a physical MAC address before local network communication.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round-Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

AIM:

To develop a simple network consisting of two PCs connected through a router using Cisco Packet Tracer, configure the required IP addresses, and verify connectivity using the ping command. To capture and analyze ICMP Echo Request and Echo Reply messages using Wireshark, examine important ICMP fields, and calculate the Round-Trip Time (RTT).

SOFTWARE / REQUIREMENTS:

Cisco Packet Tracer, Wireshark Two PCs, One Cisco 1941 Router, Ethernet cables

PROCEDURE:

1. Open Cisco Packet Tracer and create a network topology consisting of two PCs and one router.
2. Connect PC0 to the router and PC1 to the router using appropriate Ethernet connections. Wait until the connections become active.
3. Configure the router interfaces with suitable IP addresses belonging to two different networks. Enable the interfaces using the no shutdown command.
4. Configure the IP address, subnet mask, and default gateway of PC0 according to the network connected to the first router interface.
5. Configure the IP address, subnet mask, and default gateway of PC1 according to the network connected to the second router interface.
6. Verify the router interface configuration using the command:

```
show ip interface brief
```

Ensure that the required interfaces are in the up/up state.

7. From PC0, use the ping command to test connectivity with the router and then ping PC1. Verify that successful Echo Replies are received.

8. Open Wireshark on the computer and select the appropriate interface. For the actual ICMP capture performed in this experiment, the Loopback (lo0) interface was selected.

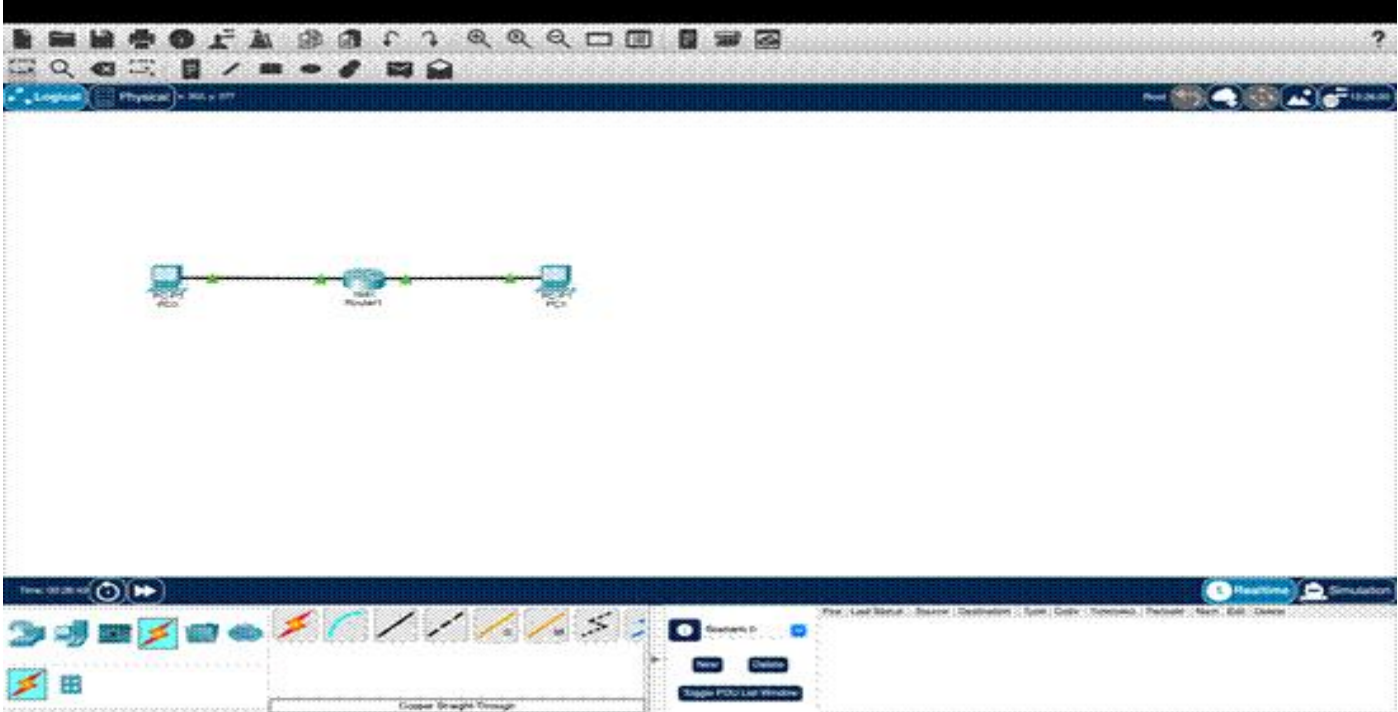
9. Start the Wireshark capture and generate ICMP traffic using the Terminal command:

```
ping -c 4 127.0.0.1
```

After four requests are generated, stop the capture and apply the display filter: icmp

10. Examine the captured ICMP Echo Request and Echo Reply packets. Record the Type, Code, Identifier, Sequence Number, and response time, and calculate the RTT from the captured timestamps.

Network Topology



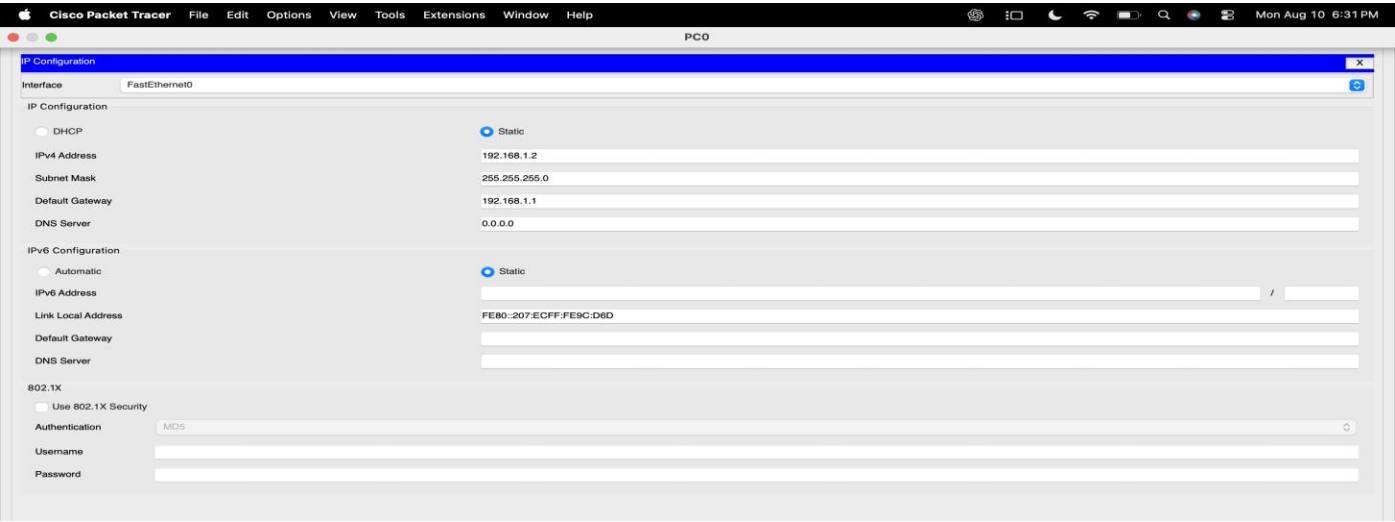
The topology consists of two PCs connected to a central router. The two PCs are configured in different IP networks, with the router providing communication between them. The connections are established successfully between the devices.

Router Configuration:



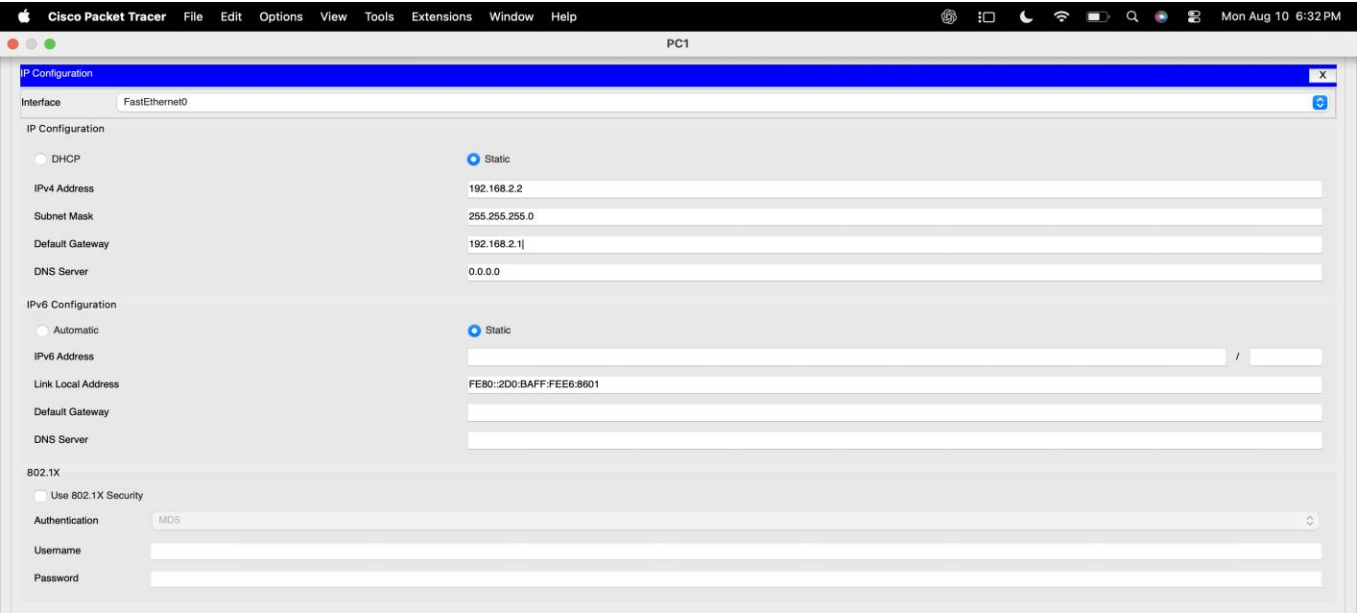
The router interfaces are configured with IP addresses belonging to the two different networks. The show ip interface brief command is used to verify the configuration and interface status. The interfaces are shown as up/up, indicating that they are operational.

PC0 Configuration:



PC0 is assigned an IP address in the first network along with the required subnet mask. The router's first interface is configured as the default gateway. This allows PC0 to communicate with devices outside its local network.

PC1 CONFIGURATION:



PC0 to Router Connectivity:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The ping command is used to verify communication between PC0 and the router. Successful Echo Replies are received from the router interface. This confirms that the IP configuration and local network connection are functioning correctly.

PC0 to PC1 Connectivity:

```
C:\>ping 192.168.2.2

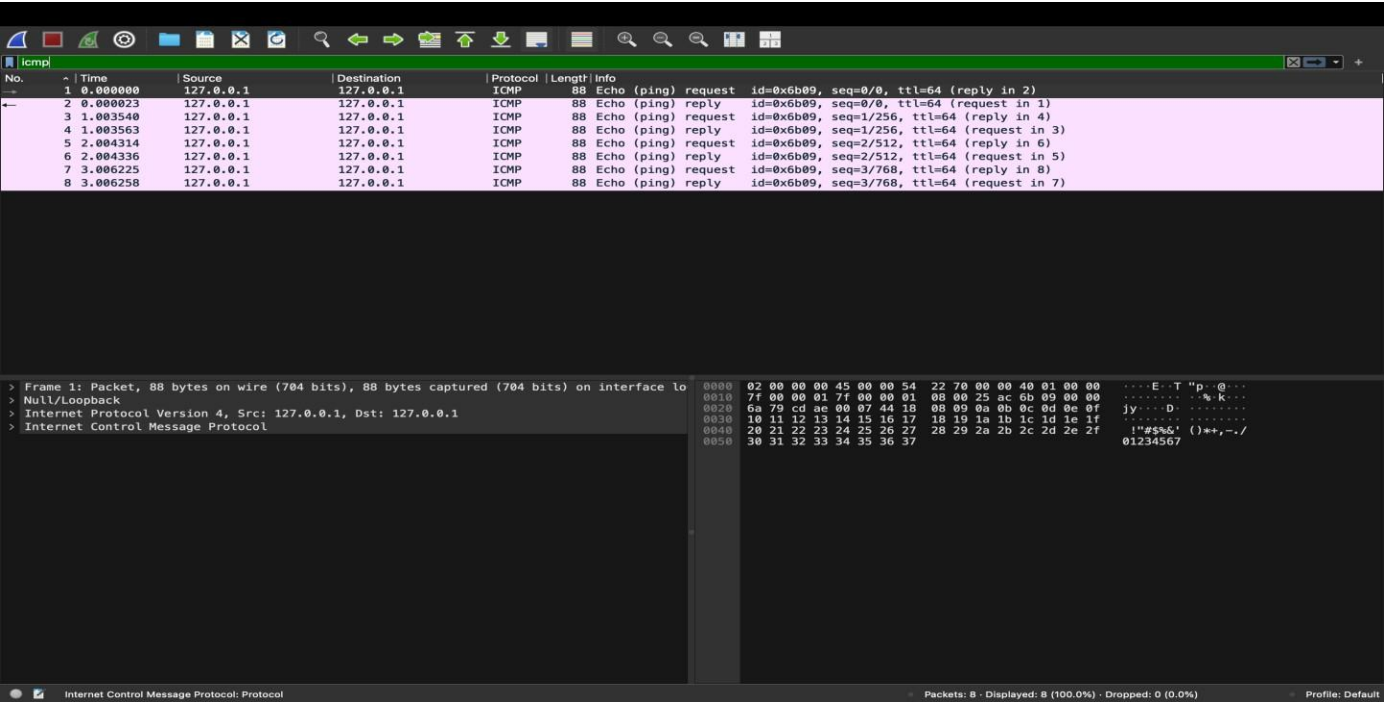
Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

WIRESHARK OUTPUT:



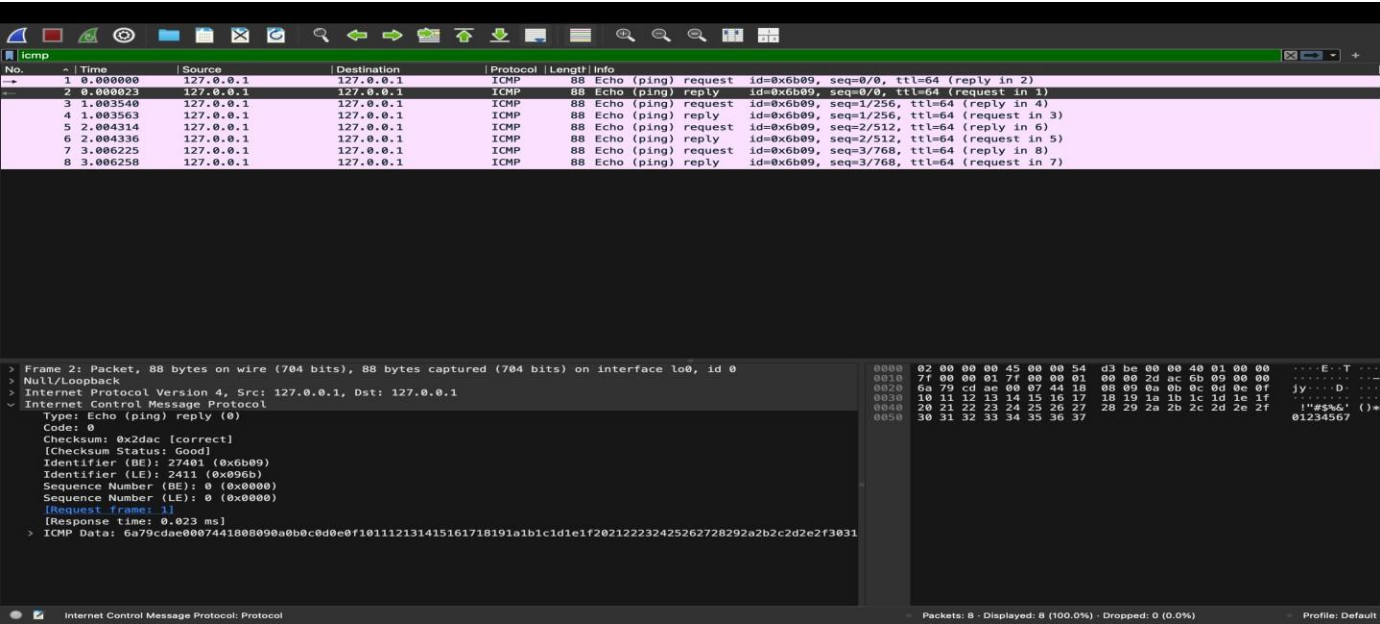
The Wireshark capture contains ICMP Echo Request and Echo Reply packets generated by the ping operation. The icmp display filter is applied to display only ICMP traffic. Four Echo Request and four corresponding Echo Reply packets are visible in the capture.

ICMP Echo Request Analysis:

```
> Frame 1: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface lo0, id 0
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0x25ac [correct]
  [Checksum Status: Good]
  Identifier (BE): 27401 (0x6b09)
  Identifier (LE): 2411 (0x096b)
  Sequence Number (BE): 0 (0x0000)
  Sequence Number (LE): 0 (0x0000)
  [Response frame: 2]
> ICMP Data: 6a79cdae0007441808090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637
```

The selected packet is an ICMP Echo Request with Type 8 and Code 0. The Identifier is 27401 and the Sequence Number is 0. These fields are used to identify the ICMP request and associate it with its corresponding reply.

ICMP Echo Reply Analysis:



The selected packet is an ICMP Echo Reply with Type 0 and Code 0. Its Identifier and Sequence Number match those of the corresponding Echo Request. Wireshark shows a response time of 0.023 ms for this request-reply pair.

Calculation:

For the first request-reply pair:

RTT=TReply –TRequest
RTT=0.000023–0.000000
RTT=0.000023 s
Therefore,

RTT=0.023 ms
Average RTT:

40.023+0.023+0.022+0.033 =0.02525 ms

Therefore,

Average RTT≈0.025 ms

RESULT

Thus, a simple network consisting of two PCs connected through a router was successfully created and configured in Cisco Packet Tracer. Connectivity was verified using the ping command. ICMP Echo Request and Echo Reply packets were successfully captured and analyzed using Wireshark, the required ICMP fields were identified, and the Round-Trip Time was calculated successfully.

